Experiment 13: Egg Whites — Nucleated Boiling Below 30°C and Vapor Exclusively with Flame OFF

Experimental Protocol

Container: Stainless steel pot (leiteira). **Liquid:** Egg whites with residual yolk traces. **Heat source:** Domestic gas flame at maximum intensity, largest burner. **Thermometry:** Digital probe thermometer (TP300, immersed in liquid). **Recording:** Continuous smartphone video, frame-by-frame analysis.

Complete Data: Experiment 13 (1 Collage, 6 Frames)

Coll.	Fr	Frame #	Time	Flame	T (°C)	Nucl. Boil.	Vapor	Observation
C1	F1	1	00:000	ON	29.1	++	–	Baseline. Flame ON. Nucleated boiling already active — bubbles rising to surface. T = 29.1°C. Yellow label: 'Why Have I Lost My Respect For The Academy?' Red circles mark SVEs/nucleation sites.
C1	F2	422	14:061	ON	~100	++	–	T ≈ 100°C. Flame ON. Vigorous boiling. ZERO visible vapor above surface despite being at boiling point.
C1	F3	686	22:833	ON	100.3	++	–	T = 100.3°C. Flame ON. Still ZERO visible vapor. The system is above boiling point with active flame — yet produces no visible vapor.
C1	F4	871	~29:0	OFF	99.5	+	+	Flame OFF. T = 99.5°C (dropping). Vapor becomes visible — faint but present. Vapor appears ONLY after flame removal.
C1	F5	1004	33:467	OFF	98.6	+	++	Flame OFF. T = 98.6°C. Vapor slightly more visible. Blue annotation marks vapor region.
C1	F6	1169	38:962	OFF	98.1	+	++	Flame OFF. T = 98.1°C. Vapor visible. Blue annotation marks vapor. Temperature falling — yet vapor persists and intensifies.

Photographic Documentation

Figure 55: E13c1 — NUCLEATED BOILING BELOW 30°C + ZERO VAPOR AT 100°C + VAPOR ONLY WITH FLAME OFF (6 frames). Stainless steel pot with egg whites (residual yolk), digital probe thermometer immersed. [F1, frame 1, t=00:000, flame ON, T=29.1°C]: Baseline — nucleated boiling already active with bubbles rising to surface at 29.1°C. Yellow annotation: 'Why Have I Lost My Respect For The Academy?' Red circles mark nucleation sites/SVEs. [F2, frame 422, t=14:061, flame ON, T≈100°C]: At boiling point with active flame — ZERO visible vapor above surface. [F3, frame 686, t=22:833, flame ON, T=100.3°C]: Above boiling point, flame still ON — still ZERO visible vapor. [F4, frame 871, t≈29:0, flame OFF, T=99.5°C]: Flame extinguished — vapor immediately becomes visible. Temperature LOWER than F3 yet vapor appears. [F5, frame 1004, t=33:467, flame OFF, T=98.6°C]: Vapor slightly more visible at lower temperature. Blue annotation marks vapor region. [F6, frame 1169, t=38:962, flame OFF, T=98.1°C]: Vapor continues. Temperature falling yet vapor persists. **Summary:** Nucleated boiling begins below 30°C. Visible vapor is ABSENT at 100.3°C with flame ON and PRESENT at 98.1°C with flame OFF. Vapor visibility is controlled by flame state, not temperature.

Anomaly Analysis — Experiment 13

Anomaly E1: Nucleated boiling below 30°C. At T = 29.1°C with flame ON, vigorous nucleated boiling is already underway — bubbles actively rise to the surface. Classical nucleated boiling in water requires local superheating at nucleation sites, which at atmospheric pressure does not produce visible sustained bubble departure at 29°C. The egg-white protein matrix either lowers the effective nucleation energy barrier to near-ambient levels, or the observed structures are SVE-mediated — consistent with Experiment 9 (microscopic thermal attraction of SVEs toward heat at similar temperatures).

Anomaly E2: Zero visible vapor at 100.3°C with flame ON. At T = 100.3°C — above the boiling point at standard atmospheric pressure — with maximum flame actively heating the system, ZERO visible vapor is observed above the liquid surface (F2, F3). Classical thermodynamics predicts that at or above 100°C, the vapor pressure of water equals atmospheric pressure, and vigorous evaporation produces clearly visible steam. The complete absence of visible vapor at 100.3°C with active heating directly contradicts the temperature-threshold model of vaporization.

Anomaly E3: Vapor appears exclusively with flame OFF. Immediately upon flame extinction (F4, T = 99.5°C), vapor becomes visible. Vapor persists and slightly intensifies as temperature drops to 98.6°C (F5) and 98.1°C (F6). The system produces visible vapor at 98.1°C without flame but NOT at 100.3°C with flame. This is the inverse of every classical prediction: vapor visibility is anti-correlated with both temperature and energy input. The controlling variable is the flame state — the active thermal gradient — not the temperature.

Anomaly E4: Vapor visibility anti-correlated with temperature. The complete dataset shows: no vapor at 100.3°C (flame ON) → vapor at 99.5°C (flame OFF) → more vapor at 98.1°C (flame OFF). Vapor intensity increases as temperature decreases. This is the exact inverse of classical vapor-pressure thermodynamics, where vapor production scales monotonically with temperature. This reproduces and extends the Experiment 11 finding (vapor at 29.5°C) in a fundamentally different system — egg whites vs. herbal infusion — and adds the new observation that vapor is actively SUPPRESSED by the thermal gradient during heating.

Systematic Falsification — Experiment 13

Classical Prediction	Observation	Status
Nucleated boiling requires local superheating well above ambient	Nucleated boiling active at 29.1°C — ~71°C below boiling point	**FALSIFIED**
Visible vapor production scales with temperature (more T → more vapor)	Zero vapor at 100.3°C (flame ON); visible vapor at 98.1°C (flame OFF)	**FALSIFIED**
Vapor visibility is a temperature-threshold phenomenon	Vapor controlled by flame state, not temperature — appears only when flame is OFF	**FALSIFIED**
Active heating maximizes vapor production	Active heating SUPPRESSES vapor; vapor appears only after heat removal	**FALSIFIED**

Cross-Experiment Consistency

Pattern	Previous Experiments	Experiment 13
Vapor at sub-boiling T	Exp. 11: vapor at 29.5°C (camomile); Exp. 4: milk overflow at 82.3°C	Nucleated boiling at 29.1°C — bubble departure ~71°C below boiling
Vapor controlled by flame state, not T	Exp. 3: motion tracks flame, not T; Exp. 11: vapor onset coincides with flux transition	Zero vapor at 100.3°C flame ON; vapor at 98.1°C flame OFF

Pattern	Previous Experiments	Experiment 13
Active gradient suppresses observable phenomena	Exp. 5–6: T drops during heating (endothermic absorption); Exp. 11: 23× post-flame heating rate	Active flame suppresses vapor at 100°C; removal releases it at 98°C
Universality across solute systems	Water (Exp. 3–4), legumes (Exp. 5–6), herbs (Exp. 11)	Egg whites — fourth distinct solute system exhibiting flame-state-dependent anomalies

Experiment 13 introduces a critical new observation: the active thermal gradient does not merely fail to produce vapor — it actively SUPPRESSES it. At 100.3°C with flame ON, the system channels all energy into the thermal force field (endothermic storage + mechanical work), leaving no excess for visible phase transition. Upon flame removal, stored energy releases exothermically and the suppression mechanism ceases — vapor appears immediately at a lower temperature. This is the vapor-domain counterpart of the temperature-domain anomaly in Experiments 5–6 (temperature drops during heating) and provides the most direct evidence yet that the thermal gradient, not the temperature, is the controlling thermodynamic variable.

References

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